

RESULTS

The output of the SafeSpin – Anti Suicide Fan system is designed to provide immediate response and alert when an abnormal downward force is detected on the ceiling fan rod. The system operates continuously and ensures that appropriate actions are taken as soon as a dangerous condition is identified.

Under normal operating conditions, the system remains inactive, and the ceiling fan functions without any interruption. The clamp and roller limit switch do not interact due to the maintained air gap, ensuring that normal vibrations do not trigger the system.

This guarantees stable and error-free operation during regular use. When an abnormal force is applied to the fan rod, axial displacement occurs, causing the clamp to press the roller limit switch. Once the switch is triggered, a signal is sent to the ESP32 microcontroller, which processes the input and activates the output mechanisms.

The primary output of the system is the activation of a buzzer, which generates an audible alert to attract the attention of nearby individuals. This immediate sound alert helps ensure that people in the vicinity can quickly respond to the situation. In addition to the local alert, the ESP32 sends a real-time notification through the Telegram messaging platform using Wi-Fi connectivity.

This alert is delivered instantly to predefined contacts such as wardens, caregivers, or family members, enabling remote monitoring and quick intervention. Overall, the system output ensures fast detection, immediate alert generation, and effective communication, thereby enhancing safety and increasing the chances of timely assistance.

